

HARI JOSHI

Kathmandu, Nepal

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RESEARCH INTERESTS

Perception and local planning for small aerial robots, when the onboard computer is limited. Arrival estimation from sparse GPS, and reliable location tracking on mobile clients, when the network link drops.

EDUCATION

Institute of Engineering (IOE), Thapathali Campus

May 2021 – May 2025

Bachelor of Electronics, Communication and Information Engineering

Kathmandu, Nepal

- **Percentage:** 63.99%
- **Relevant Coursework:** Control Systems, Microprocessors & Microcontrollers, Artificial Intelligence, Digital Signal Processing, Computer Networks, Operating Systems, Object-Oriented Programming, Embedded System Design.

RESEARCH & CAPSTONE PROJECTS

Vision-Based Autonomous UAV for Obstacle Detection, Avoidance, and Navigation | *Capstone* 🌐 **2024 – 2025**

- **Team & role:** Final-year major project at IOE, Thapathali Campus. Team lead of a 4-member team. Supervisor: Er. Umesh Kanta Ghimire (Head of Department).
- Built an F330 quadrotor (Pixhawk 4X, Raspberry Pi 4B, Intel RealSense D435) that detects obstacles with YOLOv8, quantized through ONNX and TFLite (FP16/INT8) and run with OpenCV DNN, then replans with an occupancy map and RRT* on the Pi, sending commands to the Pixhawk over UART.
- Flew the stack in Gazebo, then in the field: a college parking lot, a forest area, and open ground.
- On a logged flight the auto segment was 0–1 m/s (average 0.36 m/s) at about 0.9–1.5 m altitude. Early flights were limited by GPS interference from the Pi and the battery; raising the GPS module reduced it.

Public Transportation Assistance using Artificial Neural Network | *Minor Project* 🌐

2024

- **Team & role:** Third-year minor project at IOE, Thapathali Campus. Team lead of a 4-member team. Supervisor: Er. Kiran Chandra Dahal.
- Built an Arduino Mega tracker (NEO-6M GPS, SIM900 GSM) that posts a fix to ThingSpeak every 30 seconds, and a Django backend with WebSockets that streams positions, routes, and fares to a Leaflet map.
- Trained a feedforward network (hidden layers of 64 and 32 units, dropout 0.3) on 2022–2023 corridor GPS records. The project report measured $R^2 = 0.965$ and MAPE 6.74% on a held-out test set.

INDUSTRY EXPERIENCE

Mobile Engineer

Jul 2026 – Present

Swift Technology

Kathmandu, Nepal

- Built a cash-in feature for a remittance app in Kotlin and Jetpack Compose that tracks ATM cash deposits in real time using polling, coroutines, and Flows.
- Added security hardening to a legacy Java/XML remittance app: reverse-engineering protection, idle-session logout, screenshot prevention, VPN detection, and new-device detection.

Mobile Engineer

Apr 2025 – Apr 2026

Kathmandu Living Labs

Kathmandu, Nepal

- Built core features of the Baato Maps Android app, including navigation, search, and real-time location data visualization.
- Built the main map interface with the MapLibre SDK and improved rendering performance.
- Fixed bugs and refactored components, reducing the crash rate by 30% in key user flows.

Android Developer Intern

Aug 2024 – Sep 2024

Uncle Sam's Technologies

Remote

- Integrated the Stripe Android SDK to implement a secure payment flow, and built a WebView component for embedded web content.

SELECTED SOFTWARE PROJECTS

Driving License Exam Nepal | *Kotlin, Jetpack Compose, Material 3*   **2025**

- Built an offline-first exam preparation app for Nepal's driving license test; published on Google Play with 1,000+ downloads.

IPO Share | *Kotlin, Android SDK, MVVM, SQLite, Retrofit*  **May 2026**

- Built an app for IPO allotment discovery and portfolio tracking, with local SQLite caching and live results fetched through Retrofit.

TECHNICAL SKILLS

Robotics & Perception: ROS / ROS 2, PX4, Pixhawk 4X, Gazebo, RGB-D Sensing (Intel RealSense D435), Occupancy Mapping, RRT*

Machine Learning: Computer Vision (YOLOv8, TFLite), TensorFlow / Keras, scikit-learn, OpenCV DNN, pandas, NumPy

Languages: C++, Python, Kotlin, Java, Dart, C, SQL, Bash

Embedded & Hardware: Arduino Mega, Raspberry Pi 4B, NEO-6M GPS, SIM900 GSM, UART Serial Communication

Mobile: Android SDK, Jetpack Compose, Flutter, Kotlin Multiplatform (KMP), Compose Multiplatform (CMP), Wear OS, MapLibre SDK, Room, Retrofit

Testing & DevOps: Automated Testing, CI/CD

Backend (Basic): Ktor, Spring Boot, Django REST Framework, Django Channels (WebSockets)

Systems & Tools: Linux, Git

HONORS & LEADERSHIP

Winner, Dristi 3.0 Hackathon | *Kathmandu Engineering College* **Jan 2025**

- Built an educational Android app teaching Nepali language, culture, and folklore to NRN children.

Secretary, Leo Club of Kathmandu-Aarambha Deurali | *Lions Club* **2024 – 2025**

- Organized a blood donation program at Thapathali Campus with 120+ donors.